

Date: 27 DEC 2024

Regd. No.

--	--	--	--	--	--	--	--	--	--

Sub Code: 23EE/EC/CS2T02

MIC23

B. Tech II Semester Supplementary Examinations, Dec-2024

CHEMISTRY

(Common to EEE, ECE, CSE)

Time: 3 hours

Max Marks: 70M

- Note:** 1. Question paper consists of two parts (**Part A and Part B**)
 2. Answer all sub questions from **Part A**
 3. Answer all questions from **Part B** with **either or choice**

PART – A			10 X 2 = 20 M		
			Marks	BTL	CO
I.	a)	Define Heisenberg uncertainty principle.	2 M	L1	CO1
	b)	Write the bond order of N ₂ molecule.	2 M	L2	CO1
	c)	Write the applications of fullerene.	2 M	L1	CO2
	d)	What is a semiconductor?	2 M	L2	CO2
	e)	Write the Nernst equation and explain its terms.	2 M	L1	CO3
	f)	Define a fuel cell.	2 M	L1	CO3
	g)	What is coordination polymerization?	2 M	L1	CO4
	h)	Write the preparation of TEFLON.	2 M	L1	CO4
	i)	Write the selection rules in IR spectroscopy.	2 M	L1	CO5
	j)	Write the principle of UV-Visible spectroscopy.	2 M	L1	CO5
PART – B			5 X 10 = 50 M		
UNIT – I					
1.	a)	What is MOT? Write its postulates.	5M	L2	CO1
	b)	Derive de Broglie's equation.	5M	L3	CO1
(OR)					
2.	a)	Derive Schrodinger wave equation of a microscopic particle in one dimensional potential box.	5M	L2	CO1
	b)	Write Energy level diagram of CO.	5M	L2	CO1
UNIT – II					
3.	a)	What is a supercapacitor? List its applications.	5M	L3	CO2
	b)	What is CNT? Write its classification and application.	5M	L2	CO2
(OR)					
4.	a)	What is HPC? Write the applications of HPC.	5M	L2	CO2
	b)	Outline the classification, properties, and applications of Graphene.	5M	L2	CO2
UNIT – III					
5.	a)	Find the concentration of Cu ²⁺ solution in the given electrochemical cell: Mg/Mg ²⁺ (0.1 M) // Cu ²⁺ (M1)/Cu Given: E ⁰ _{Mg+2/Mg} = -1.36V ; E ⁰ _{Cu+2 /Cu} = 0.34V ; E _{cell} = 1.53V at 298K.	5M	L4	CO3
	b)	What is an electrochemical sensor? Illustrate amperometric sensor with an example.	5M	L2	CO3
(OR)					
6.	a)	What is potentiometry? Explain how redox titrations are determined.	5M	L2	CO3
	b)	What is a battery? Explain the reactions involved in the battery that is used in mobiles.	5M	L2	CO3
UNIT – IV					
7.	a)	Write the differences between addition and condensation	5M	L2	CO4

		polymerization.			
	b)	Write the preparation, properties and application of Buna-S.	5M	L2	CO4
(OR)					
8.	a)	Write the preparation, properties and application of Bakelite.	5M	L2	CO4
	b)	What is a biodegradable polymer? Explain the preparation and applications of PLA.	5M	L2	CO4
UNIT – V					
9.	a)	List out the applications of UV Visible spectroscopy.	5M	L3	CO5
	b)	What is spectroscopy? List the applications of IR spectroscopy.	5M	L3	CO5
(OR)					
10.	a)	What is an electronic transition? Illustrate different types of electronic transitions in UV spectroscopy.	5M	L2	CO5
	b)	What is chromatography? Explain its classification and applications.	5M	L2	CO5

***** End of the Question Paper *****